MANCHESTER 1824
The University of Manchester

EART11100

How to Build a Planet

Week 5 The Earth in Motion: The Plate Tectonic System

The development of the Theory of Plate Tectonics caused a paradigm shift in our understanding of how our Earth works.

The Plate Tectonic System: Greek '*tekton*'- builder









Geologic events that result from interactions with the Earth's interior

The Plate Tectonic System: Greek '*tekton*'- builder







- The theory that describes the movement of plates and the forces acting on them is called the **Theory of Plate Tectonics.**
- Heat in the Earth is the driving force for plate motion.

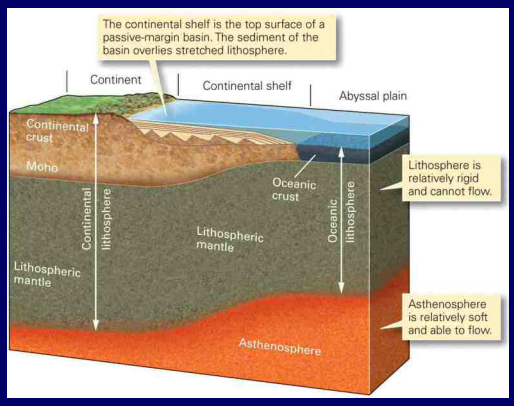
Key points of the lecture

- The Earth is differentiated into layers, not only according to their chemistry, but also based on material strength.
- The Earth's shell is fragmented into plates that are constantly moving.
- The boundaries between the plates are classified based on observations of their relative motions and geologic activity.
- The motion of plates over the Earth has led to the creation and destruction of supercontinents through time and this has impacted on global climate, sea level and the biosphere (Earth system).

Zoning of the Earth based on Material Strength

Strength: how much Earth material can resist deformation

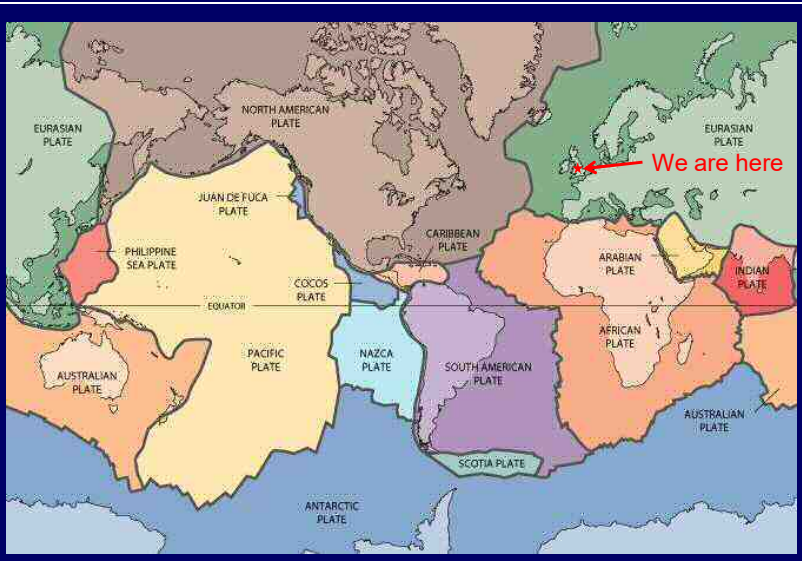
- depends on chemical composition and temperature



Lithosphere
- Greek *lithos*: 'stone'
- Cool, strong, outer shell
- Crust + ~top 100km of mantle
- kms - 200km thick
- Behaves like a relatively rigid, brittle shell

Asthenosphere
- Greek *asthenes*: 'weak'
- Hot, weaker interior
- Mantle below lithosphere
- ~300km thick
- Flows like a ductile solid

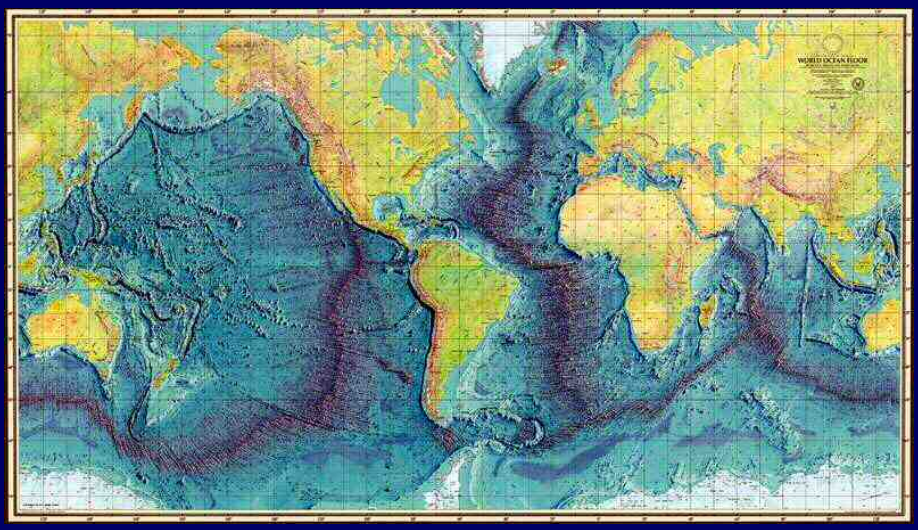
The Earth's Lithosphere is Fragmented!



Earth's surface is a mosaic of rigid plates

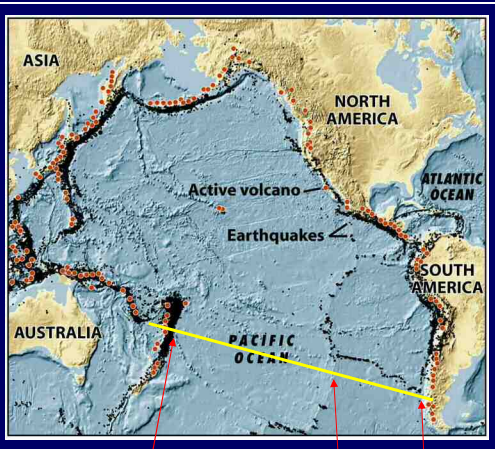
The lithospheric plates move over the Earth's surface at rates of a few cms per year

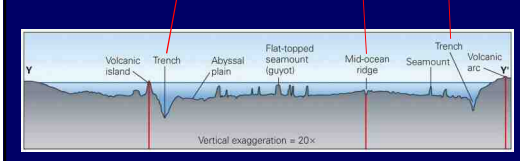
The Plates and their Boundaries

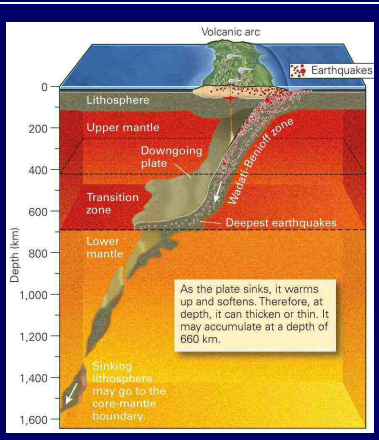


The plates are bound by mid-ocean ridges and deep-ocean trenches

Distribution of Earthquakes and Volcanoes



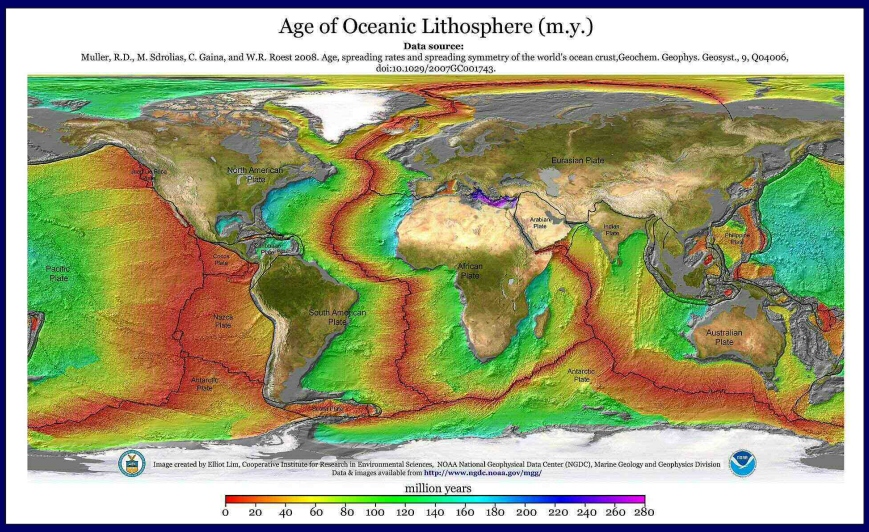




Wadati-Benioff zones:

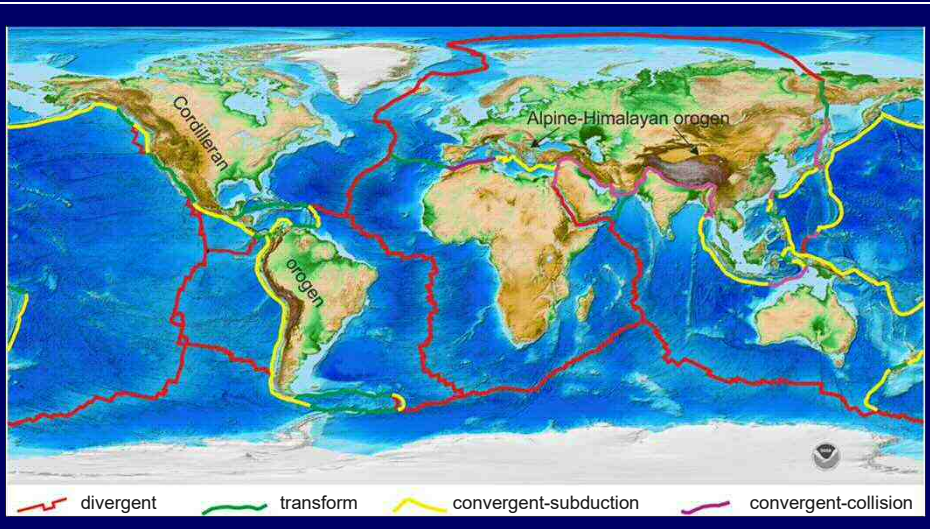
- earthquake zones parallel to trenches
- inclined 40-60°
- extend several hundred km into Earth

The Composition and Age of the Crust on Plates



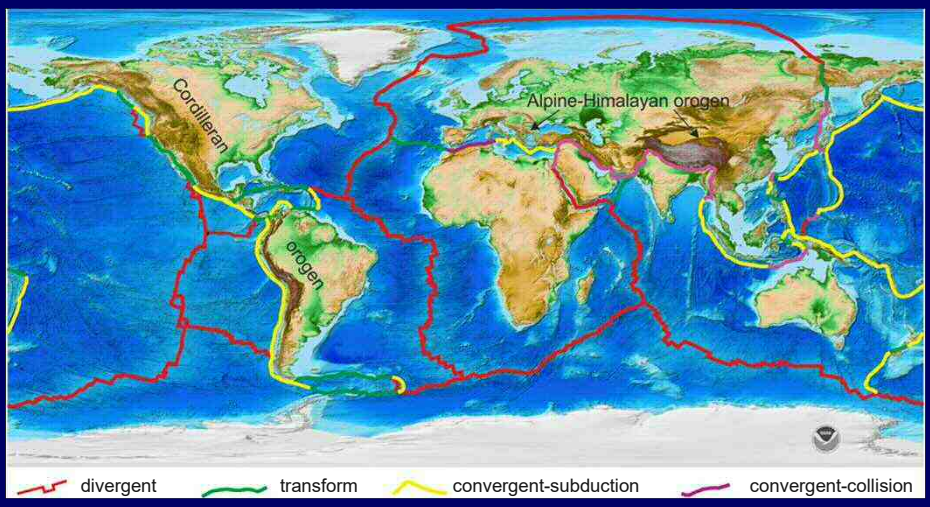
Distribution of oceanic and continental crust

The Plates and their Boundaries



Present-day Earth global relief showing distribution of divergent, convergent and transform boundaries

The Plates and their Boundaries



Plates move and change size through time

Divergent (move apart), convergent (come together), transform (slide past)

The Plates and their Boundaries

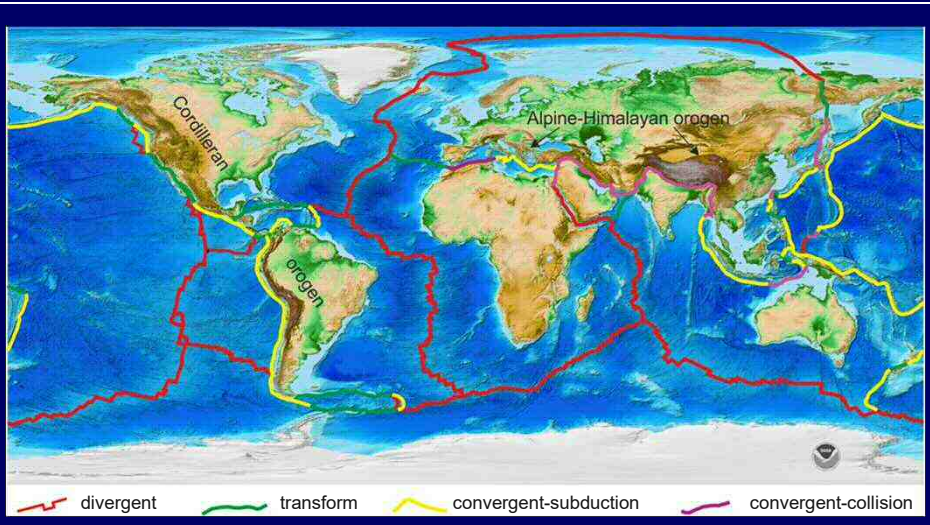
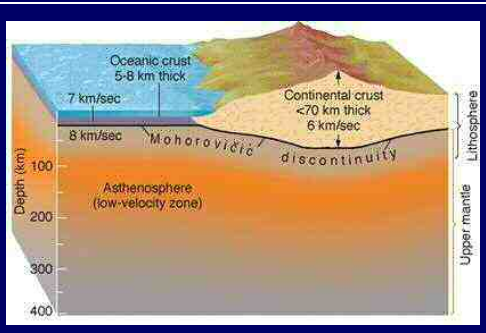


Plate boundaries are typically combinations of three types

Continental and Oceanic Lithosphere Behave Differently





Continental crust:
- underlies continents
- thicker and less dense
- av. thickness 40 km
- seismic wave velocities ~6 km/s
- composition more felsic: granite-like
- derived by a sequence of melting processes

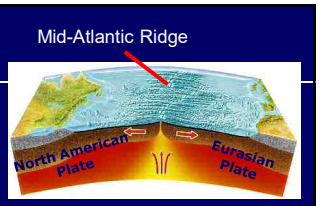


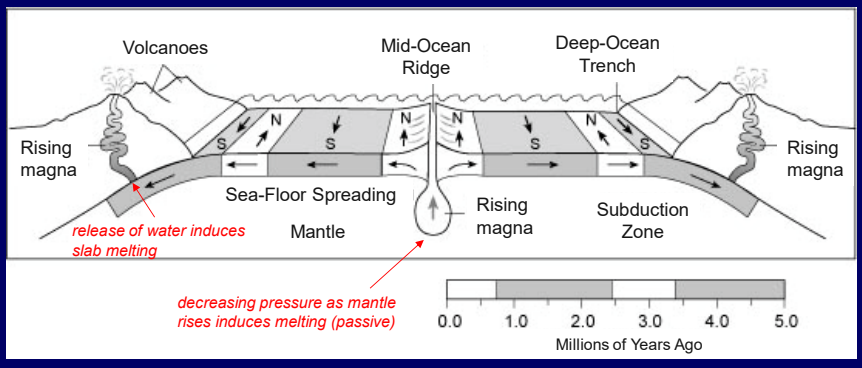
Oceanic crust:
- underlies sea floor
- thinner and denser
- av. thickness 7 km
- seismic wave velocities ~7 km/s
- composition mafic: basalt and gabbro
- result of mantle melting, deeper levels

Key differences between continental and oceanic crust at plate boundaries

Divergent or Constructive Boundaries

(i) Oceanic plate separation





New rock created at the ridges, old rocks recycled by the trenches

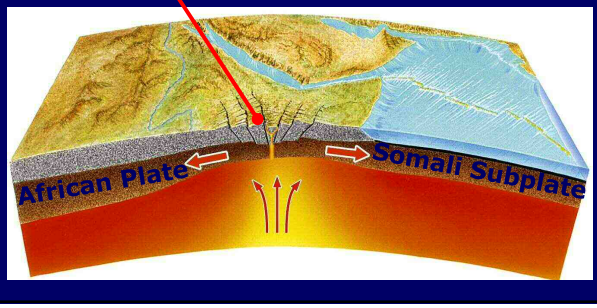
Divergent Boundaries

(ii) Continental plate separation

Rift valleys, volcanoes, and earthquakes

breaking of crust = faulting







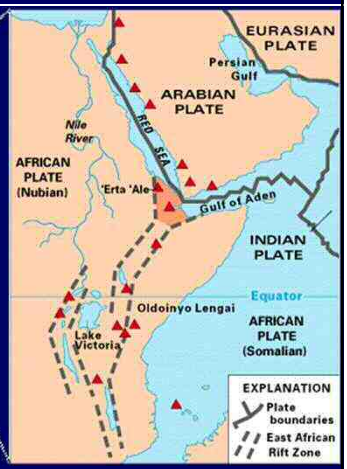
Divergent Boundaries

(ii) Continental plate separation

If separation continues, oceanic crust can form





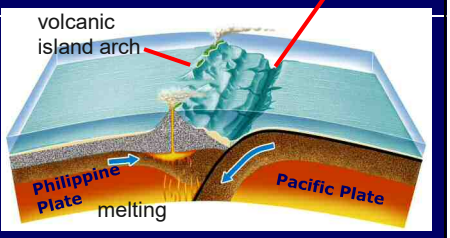


Will the East African rift valley eventually open to form a new ocean or fail?

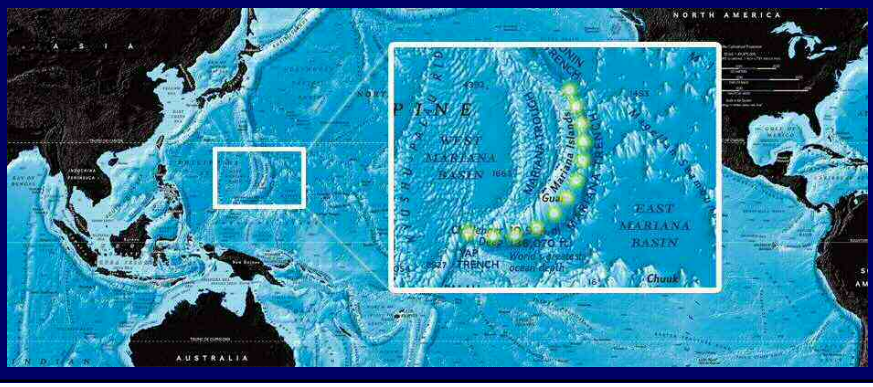
Convergent or Destructive Boundaries

(i) Ocean-ocean convergence

Subduction: one plate descends beneath the other



Deep earthquakes 690km depth



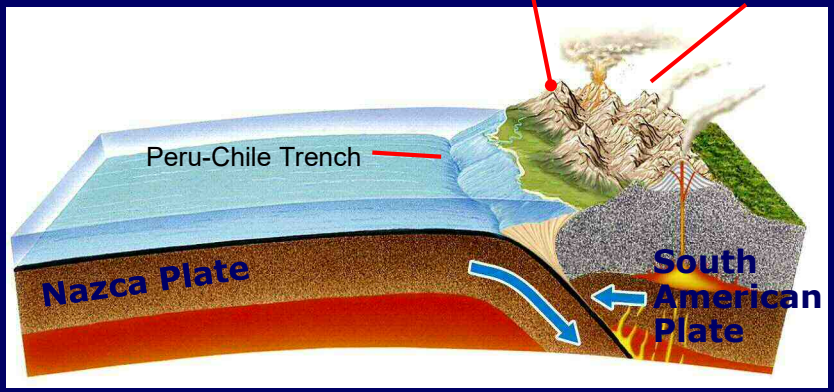
Convergent Boundaries

(ii) Ocean-continent convergence

Volcanic mountain chain, folded and thrusted mountains, deep earthquakes

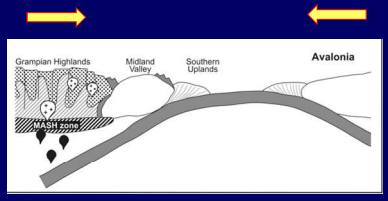


Andes Mountains

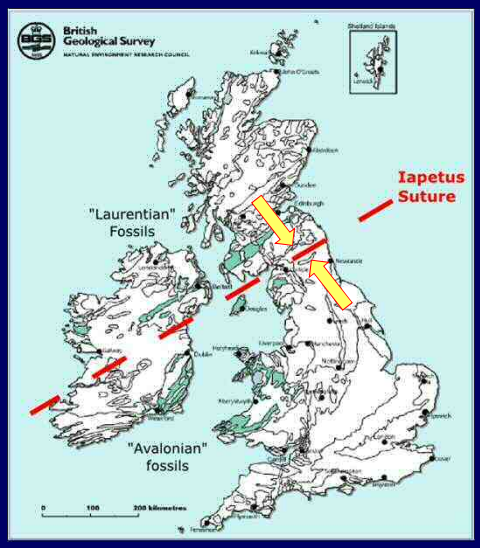


Convergent Boundaries

(ii) Ocean-continent convergence



Subduction 450-400 Ma



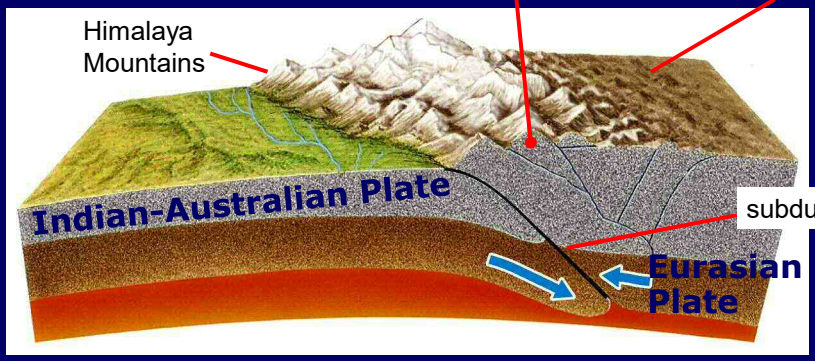
Convergent Boundaries



(iii) Continent-continent convergence

crustal thickening, folded and thrusted mountains, and earthquakes

Tibetan Plateau



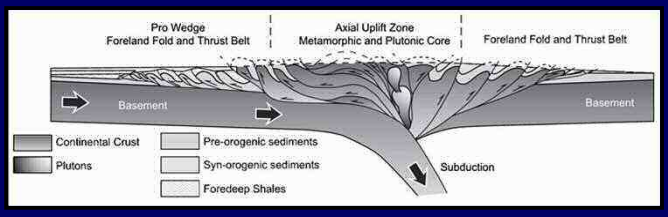
The combined processes of collision, deformation, metamorphism and intrusion of granites are known as mountain building or orogenesis

Convergent Boundaries

(iii) Continent-continent convergence



Folding associated with the Lewis Thrust, Rocky Mountains, USA



Conceptual model of an orogenic belt produced in a continent-continent collision.

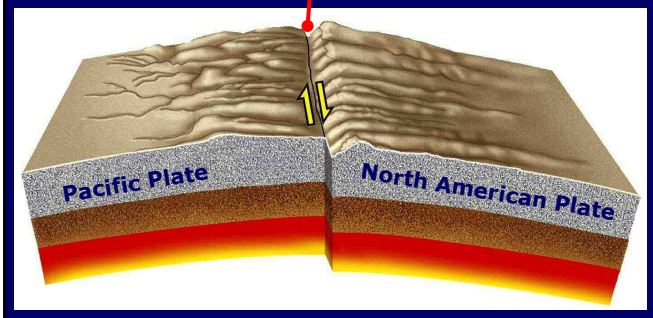
This looks much like a N-S section across the Pyrenees (Spain left; France right)

Transform-Fault Boundaries



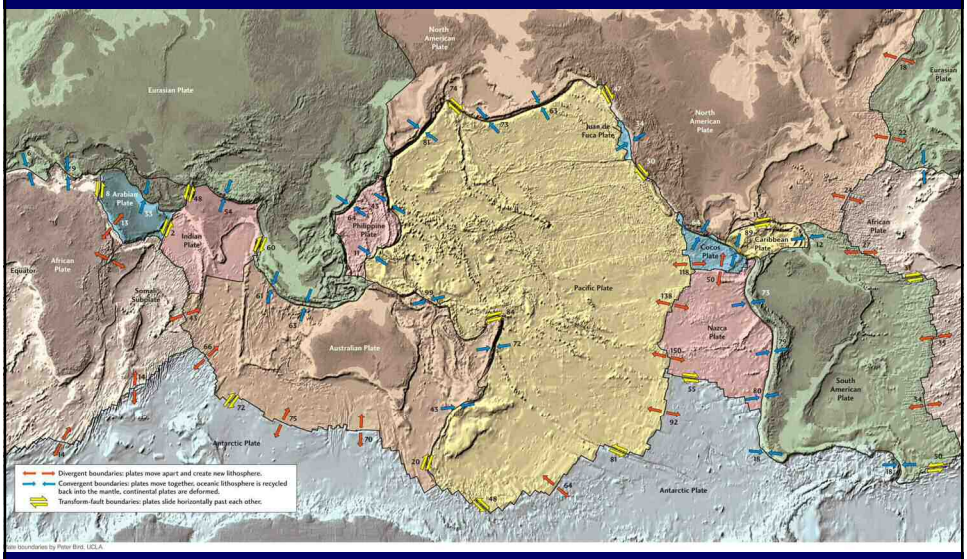
(i) Continental transform fault

Lateral (transform) fault and earthquakes



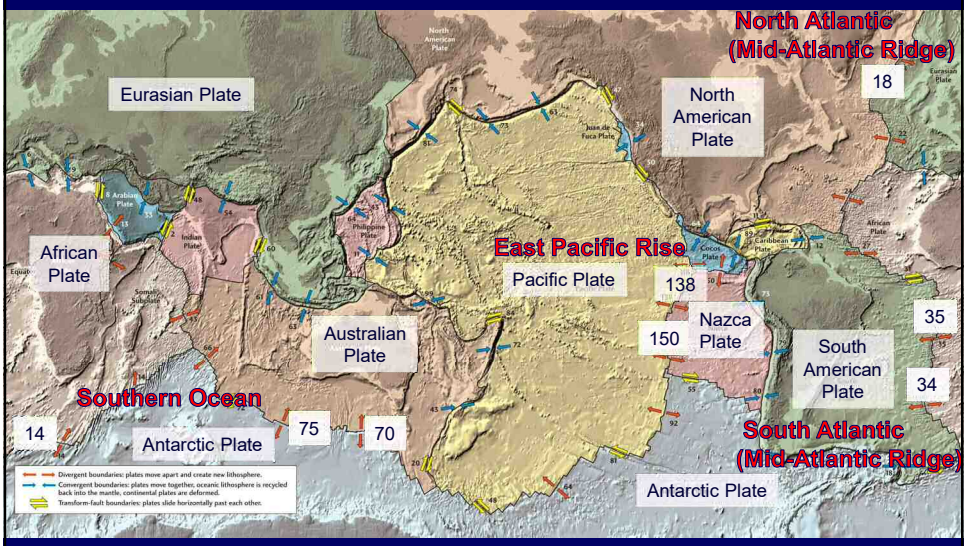


Rates and History of Plate Motion



At what rates do the plates move?

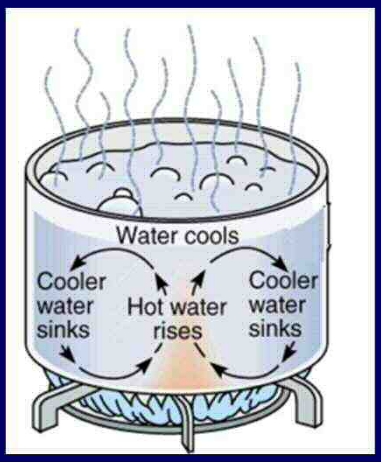
Rates and History of Plate Motion



Example relative plate velocities (mm/yr)

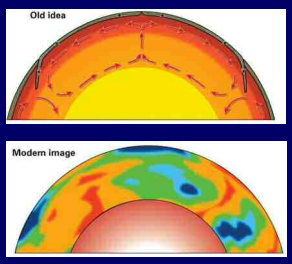
What Drives Plate Tectonics?

Plate motion is driven by mantle convection



Common analogy for mantle convection

Is this the way plates move; passively dragged along by convection currents in the mantle?

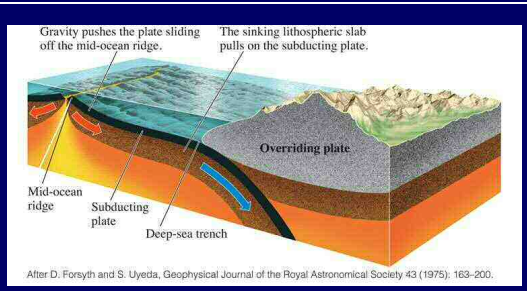


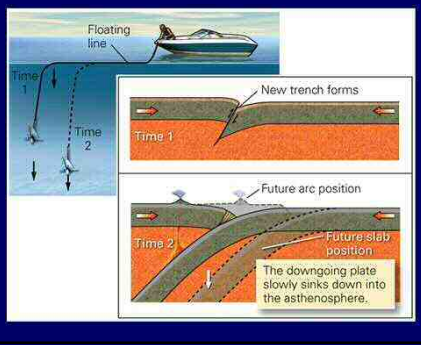
Old and modern images of mantle convection

Two forces believed to move plates:

- Ridge push
- Slab pull (including basal drag)

What Drives Plate Tectonics?

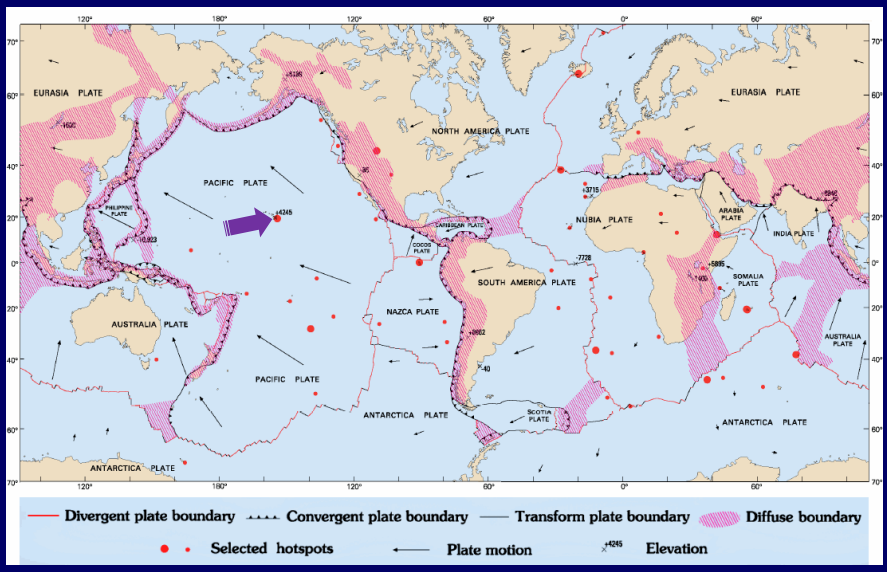




Slab pull: plates descend into the mantle under their own weight
(cold, dense plates into warm, less dense asthenosphere)

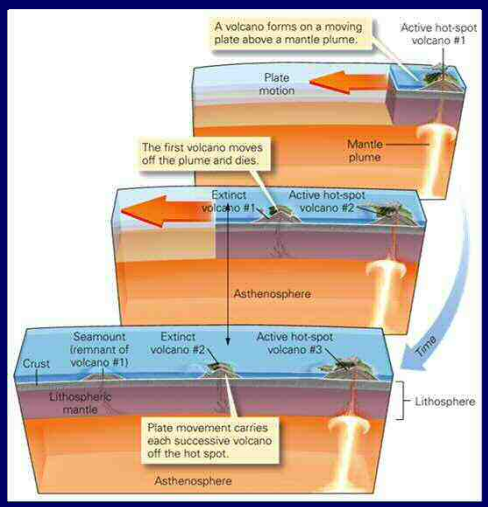
Ridge push: the plate over the mid ocean ridge is elevated, leading to gravitational sliding

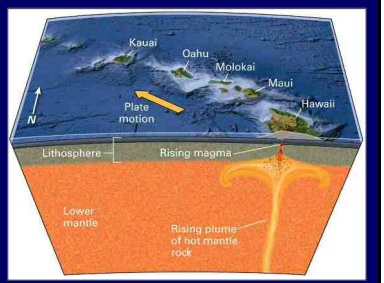
Mantle plumes and hot spots



Mantle plumes and hot spots

Hot-spot volcanoes are probably a consequence of mantle plumes (areas of hot, upwelling mantle).



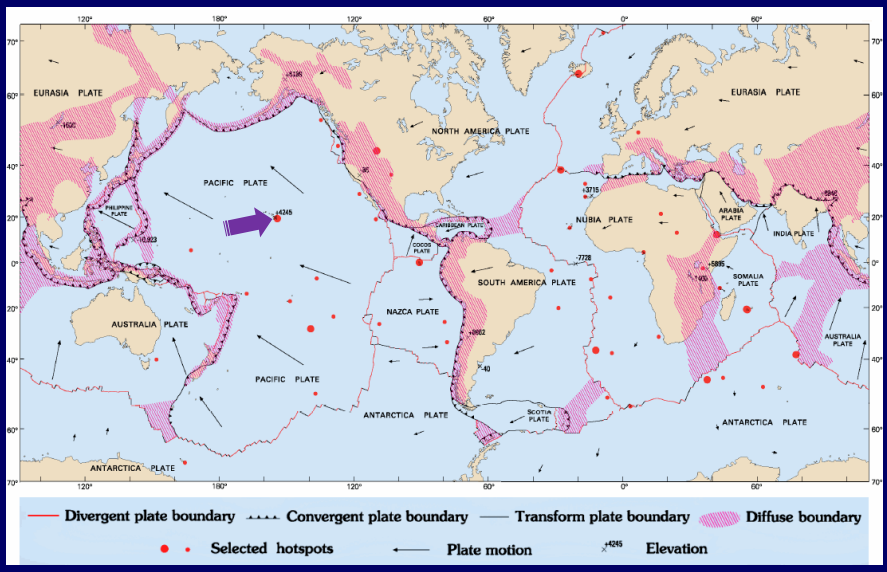


As a plate moves over a hot spot a chain of volcanoes is left behind

The Hawaiian Islands are an example of such a hot-spot track

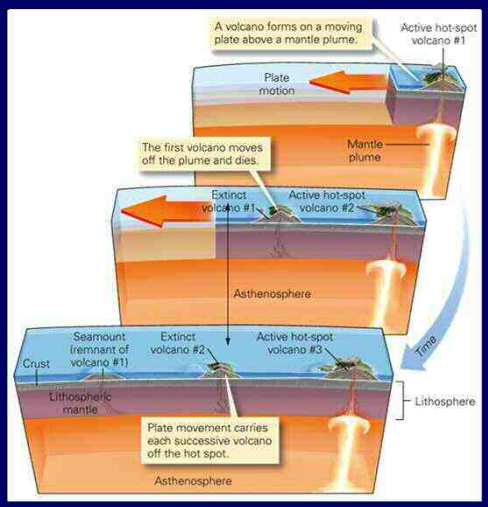
The islands increase in age with distance from the present hot spot

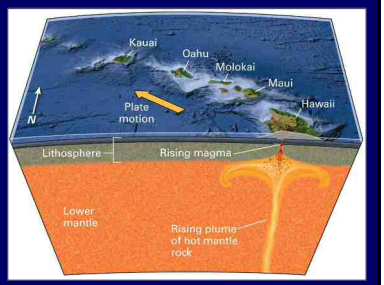
Mantle plumes and hot spots



Mantle plumes and hot spots

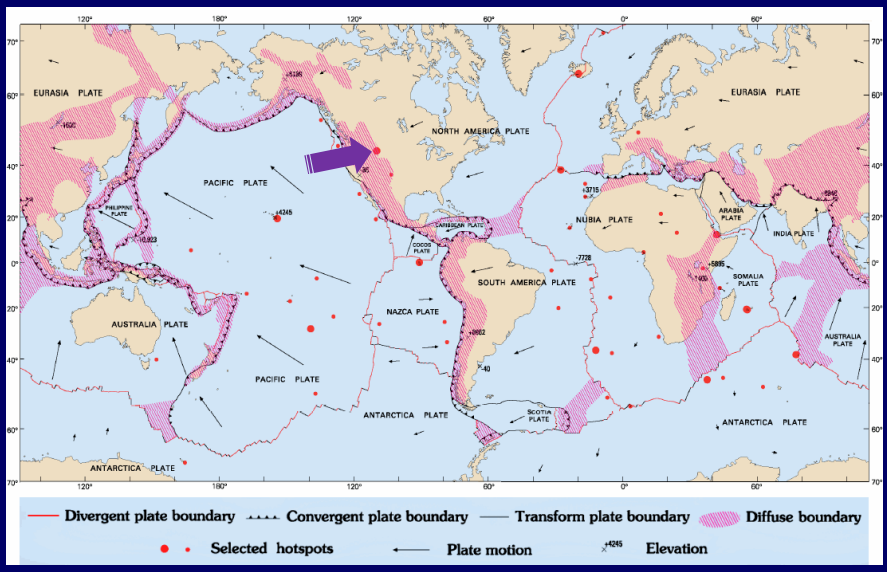
Hot-spot volcanoes are probably a consequence of mantle plumes (areas of hot, upwelling mantle).





The Hawaiian hotspot
- active 70 million years
- chain extends 6000 km

Mantle plumes and hot spots



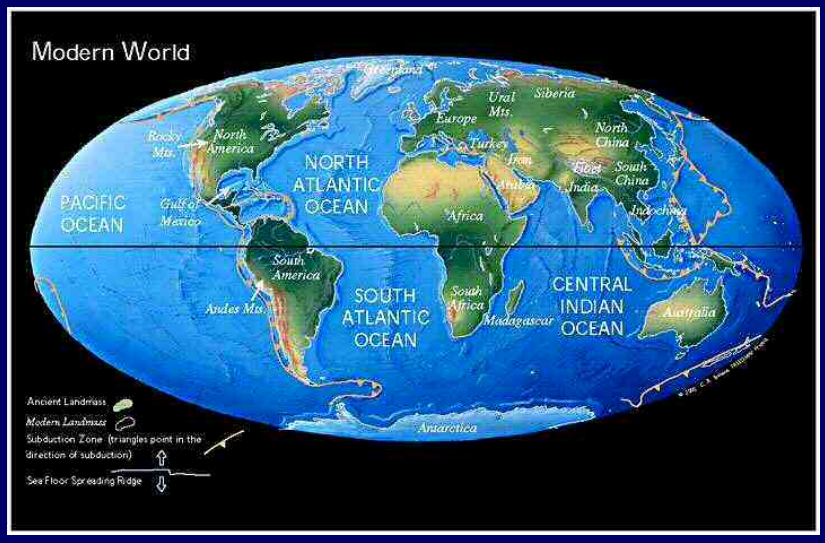
Mantle plumes and hot spots



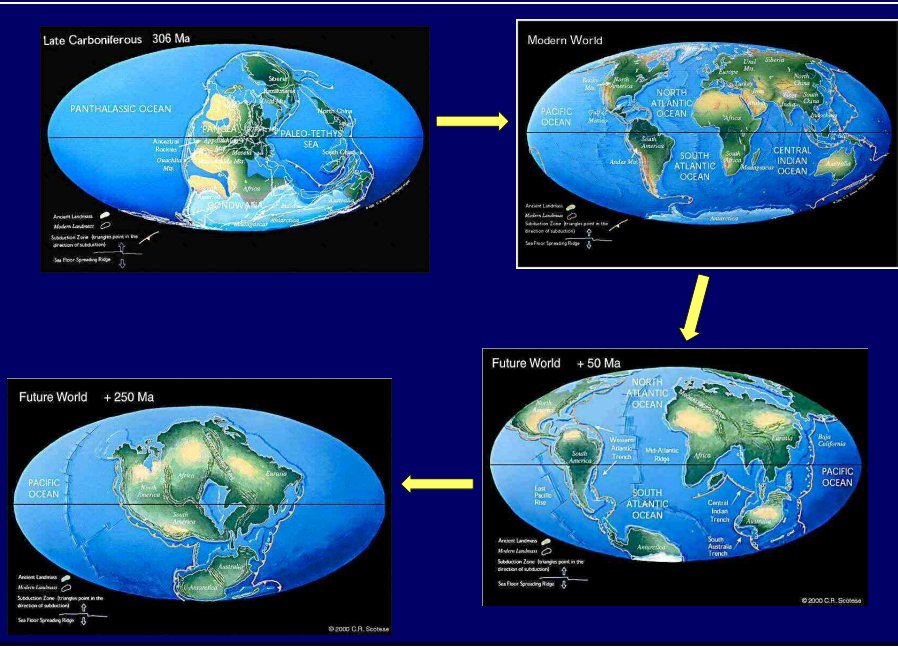
Yellowstone hotspot
- active 15 million years
- chain of calderas and volcanic features extends 650 km



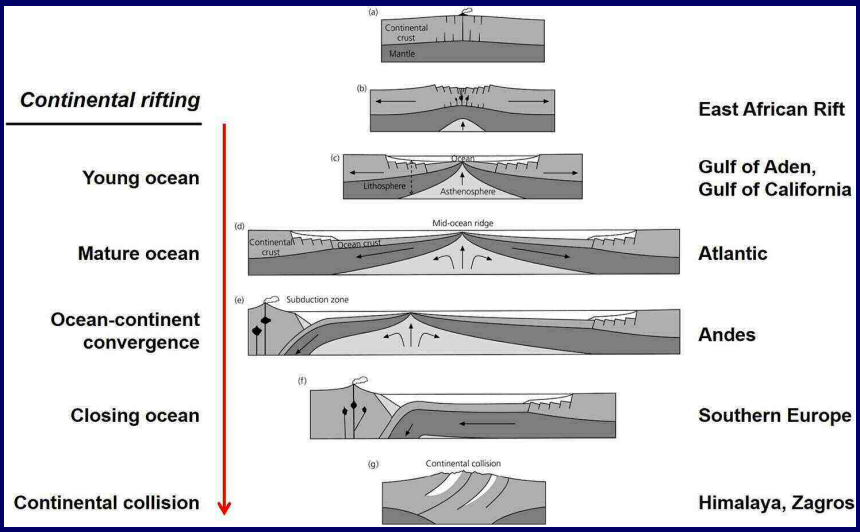
One theory to unify them all…..



Constantly on the move

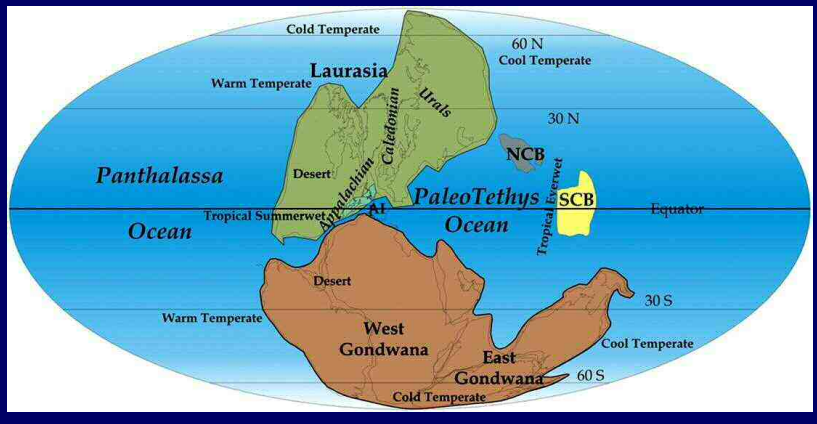


The Wilson Cycle



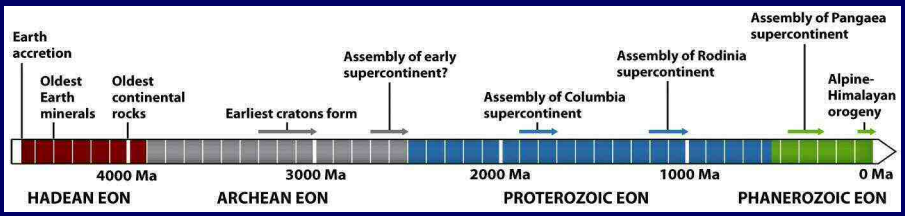
Schematic illustration of the Wilson cycle, showing modern areas at each stage
(Stein and Wysession, 2003)

The Supercontinents



- Wegener's Pangea is just the most recent in a series of supercontinents that have punctuated Earth history for billions of years
- Supercontinents: large continents that include most of all of the existing continents

The Supercontinent Cycle

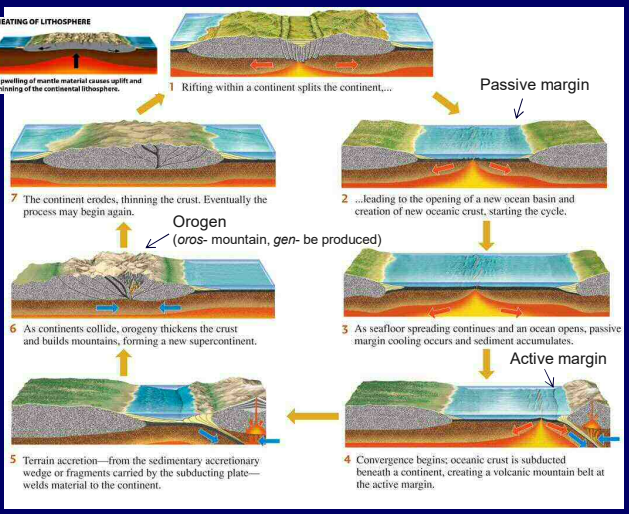


Geologic Time Scale of Events in the History of Earth's Continents

Supercontinent assembly and breakup is episodic

Debate over regularity of episodicity i.e. if this constitutes a 'cycle'

Supercontinent assembly and break up



The Wilson Cycle describes the plate tectonic processes responsible for the formation and break up of supercontinents and the opening and closing of ocean basins

Break up requires rifting e.g., mafic dyke swarms

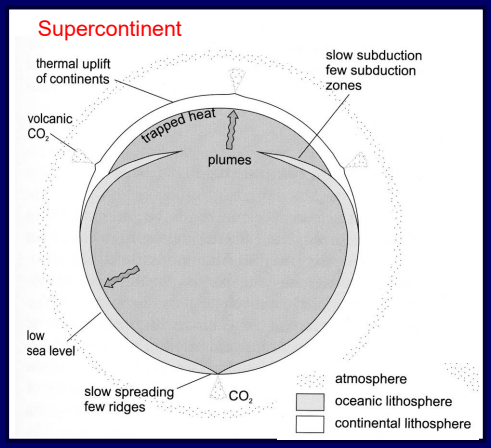
Assembly requires collision → orogeny

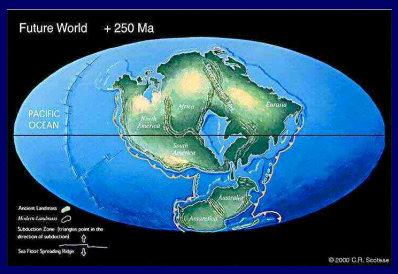
Orogeny: episodes of mountain building-processes of folding, faulting, terrain accretion, magmatic addition and metamorphism

- Oceanic-Continental
- Continental-Continental (most intense)

Oceanic-Oceanic: subduction not orogeny

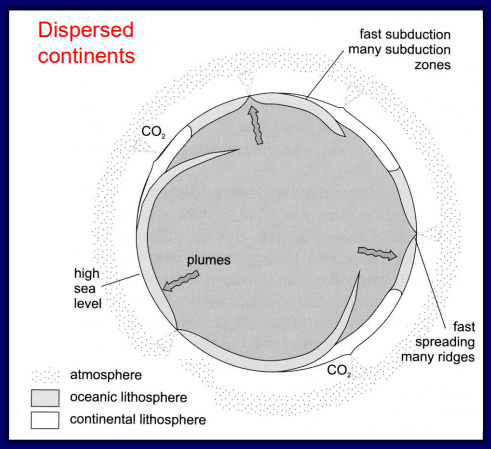
Supercontinents, global climate and sea level

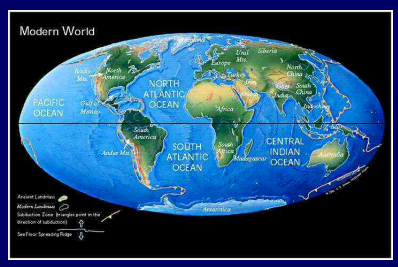




Relatively short length of mid-ocean ridges and subduction zones

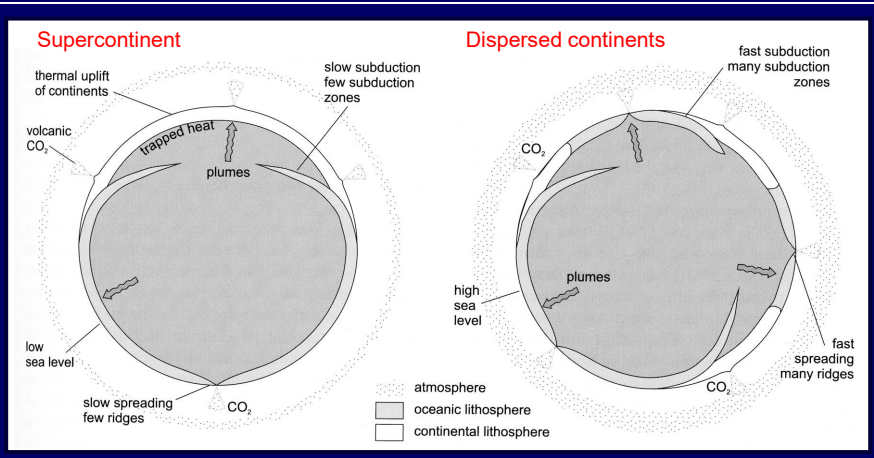
Supercontinents, global climate and sea level





Relatively long length of mid-ocean ridges and subduction zones

Supercontinents, global climate and sea level

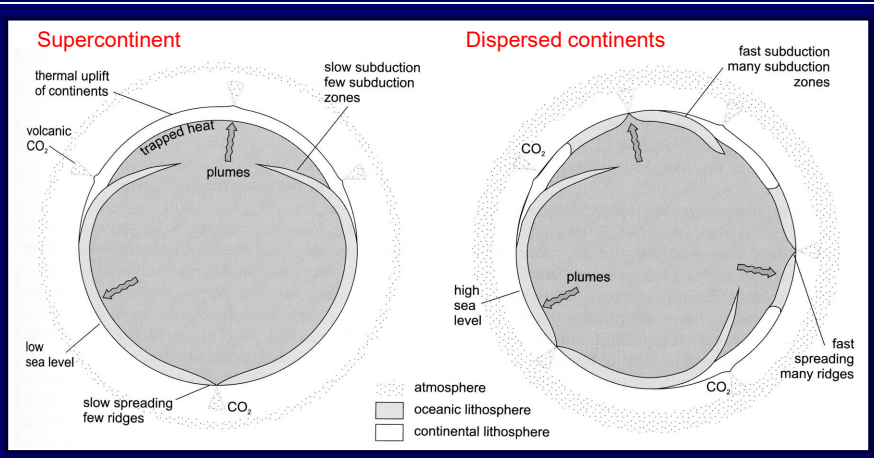


Relatively short length of mid-ocean ridges and subduction zones

→ more heat stored in the Earth and so continents stand high → low sea level

Relatively long length of mid-ocean ridges and subduction zones → less heat stored in the Earth & displaces oceanic volume → continents sit low → high sea level → large areas of shallow marine areas → enhanced volume and diversity of biologically favoured areas

Effect of CO_2 on global climate (effect of volcanic activity)



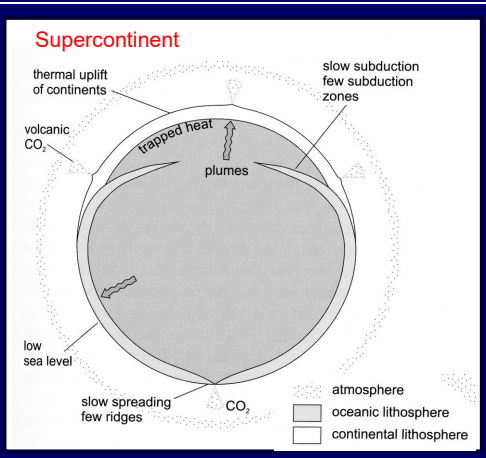
Relatively short length of mid-ocean ridges and subduction zones → low magmatic activity

→ low CO_2 supply to atmosphere (global cooling)

Relatively long length of mid-ocean ridges and subduction zones → high magmatic activity

→ high CO_2 supply to atmosphere (global warming)

Effect of CO_2 on global climate (effect of weathering)



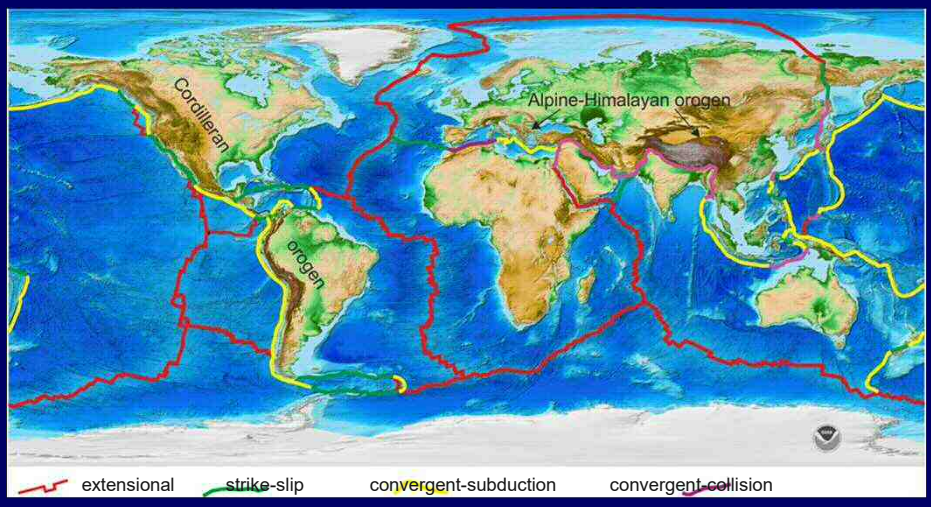
CO_2 content of atmosphere also affected by weathering processes

Mineral alteration during weathering consumes CO_2
→ burial of weathered minerals removes CO_2 from atmosphere cycle

Low global sea level
→ greater continental area for weathering
→ more CO_2 drawdown

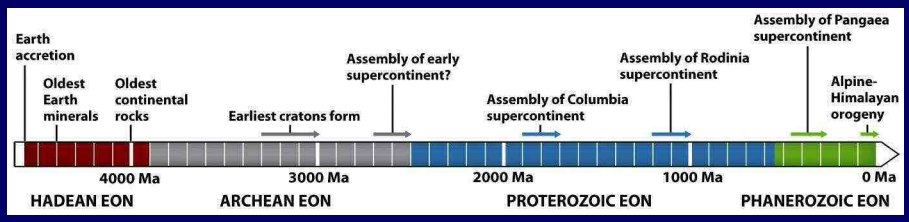
Weathering most efficient in wet, tropical zones and where there are lots of young basalts
Effect on climate from supercontinents:
→ supercontinent location matter
→ preceding magmatic history matters

Present day linked plate boundaries



How did this system evolve?

Prelude



Geologic Time Scale of Events in the History of Earth's Continents

Learning outcomes revision

Now we have completed the lecture, are you able to:

- differentiate the Earth into layers based on material strength and list the properties of each
- explain our current understanding of what drives the motions of plates today
- identify the largest tectonic plates and delineate their boundaries on a map
- differentiate divergent, convergent and transform-fault plate boundaries based on observations of relative motions and geologic activity
- explain the relationship between the Wilson cycle and the supercontinent cycle
- describe the impact that supercontinent break up and assembly has on global climate, sea level and the biosphere